Supplementary Material for Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures

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This document contains the supplemental material of the paper *Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures (S. Croci et al.)*.

I. Stimuli from the Gaussian splat vs Point cloud experiment

Figure 1 shows a sample frame from the stimuli used in the subjective experiment with the point cloud (PC) and Gaussian splat (GS) reconstructions on the left and right, respectively. The subfigure captions indicate the names of the captures used to generate the reconstructions. In the experiment, we also used the same stimuli with the left–right order of the reconstructions reversed.

II. Analysis of the Stimuli Watching Times

This section presents the analysis of the stimuli watching times collected during the subjective experiment. In particular, Figure 2 shows the histogram of the overall watching times, while Figure 3 shows the per-capture histograms for the GS vs. PC and PC vs. GS stimuli. In these histograms, there are peaks close to 10 s, i.e., the duration of the stimuli. Moreover, Figure 4 shows a bar chart of the median watching time for the GS vs. PC and PC vs. GS stimuli of each capture, with error bars indicating the interquartile range (IQR). In this chart, the medians are close to 10 s.

III. Analysis of the Dynamic Gaussian Splat Reconstructions

This section presents the GS analysis of all dynamic GS reconstructions in Figure 5, and the GS analyzes of each dynamic reconstruction in the Figures 6, 7, 8, 9, 10, and 11. As can be observed, the GS analyzes of the dynamic GS reconstructions do not vary much.

IV. Analysis of the Static Gaussian Splat Reconstructions

This section presents the GS analyzes of the static GS reconstructions of the paper [1] generated with a scene GS optimization method called Postshot [2]. Figure 12 shows the GS analysis of all static reconstructions, while Figures 13, 14, 15, 16, 17, 18, and 19 show the GS analyzes of each static GS reconstruction. It can be observed that there is a clear difference between the dynamic GS reconstructions of our paper and the static ones.

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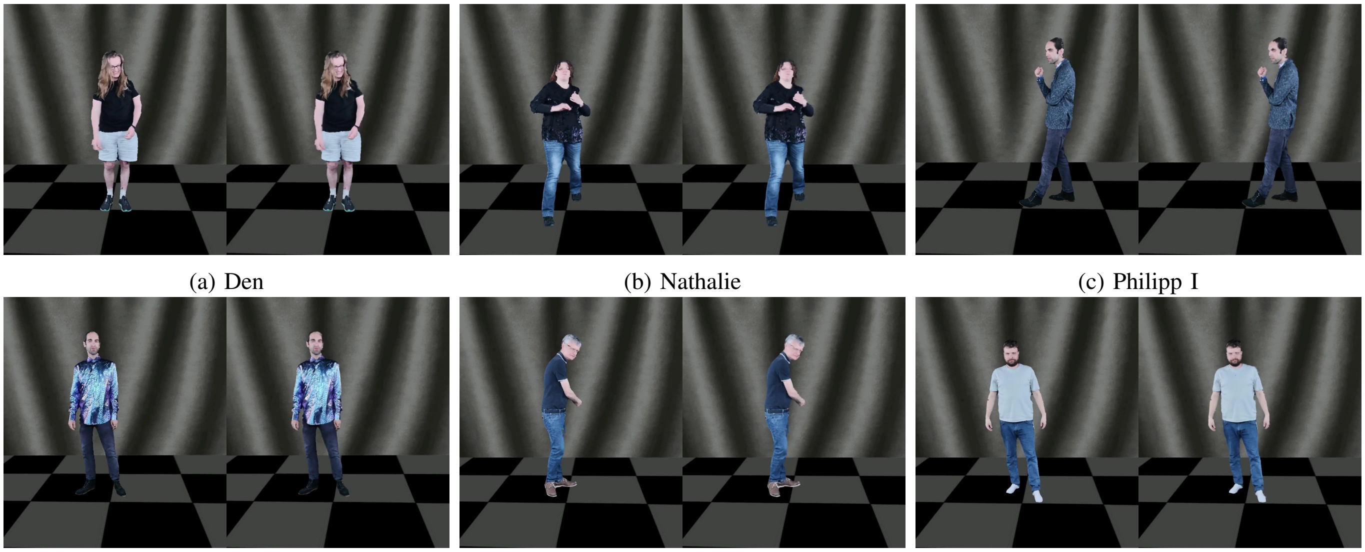
References

[1] P. Haslbauer, M. Pullen, C. Reichherzer, and A. Smolic, "Gaussian splatting vs. classical photogrammetry: A comparison for virtual backdrops," in *2025 17th International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, 2025, pp. 1–4.
[2] (2025) Postshot. [Online]. Available: https://www.jawset.com/



(a) Den (b) Nathalie (c) Philipp I

(d) Philipp II (e) Simone (f) Thanos

Fig. 1: Stimuli with point cloud and Gaussian splat reconstructions on the left and right, respectively, together with the names of the captures used to generate the reconstructions. In the subjective experiment, we also used the same stimuli with the reconstruction order reversed.

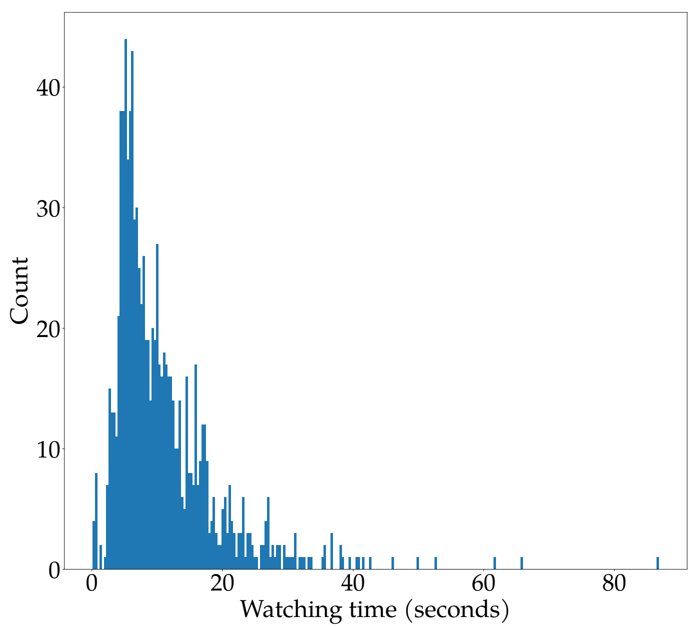
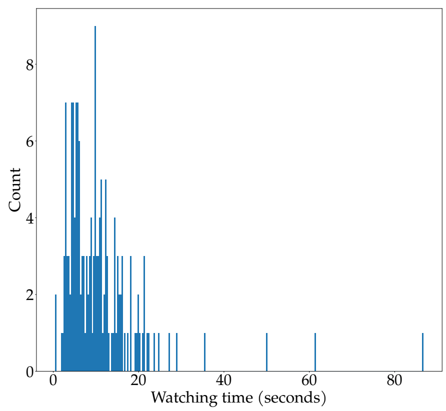
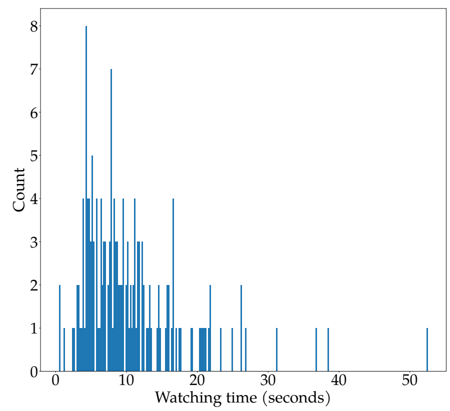


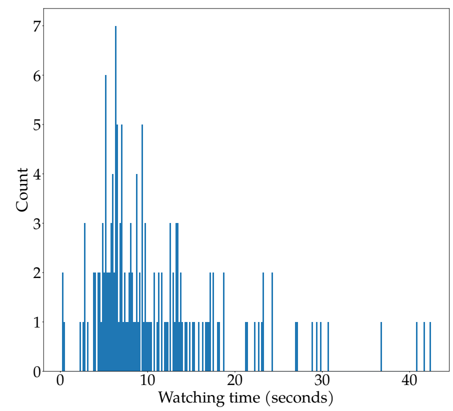
Fig. 2: Stimuli watching time histogram.



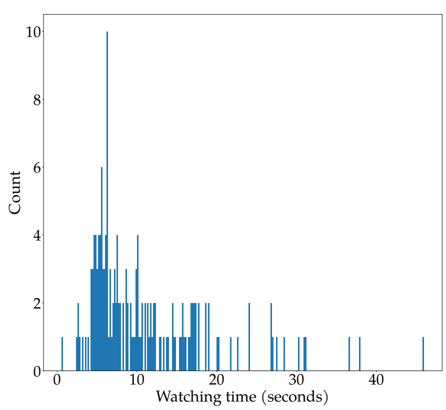
(a) Den



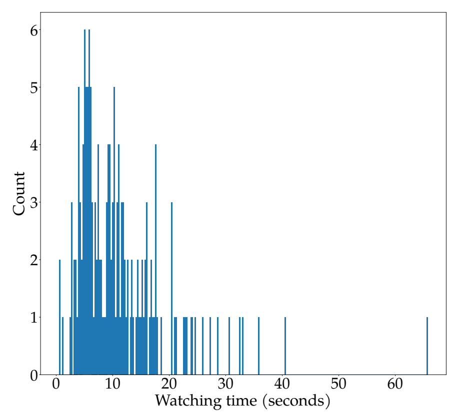
(b) Nathalie



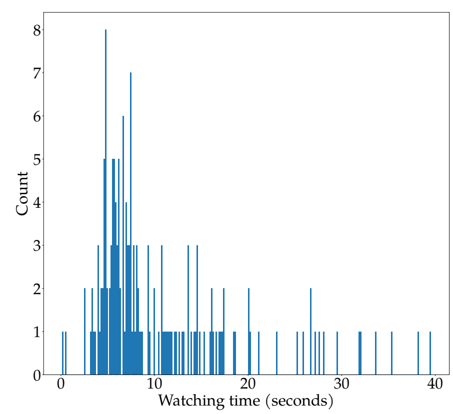
(c) Philipp I



(d) Philipp II



(e) Simone



(f) Thanos

Fig. 3: Per-capture watching time histograms for the GS vs. PC and PC vs. GS stimuli.

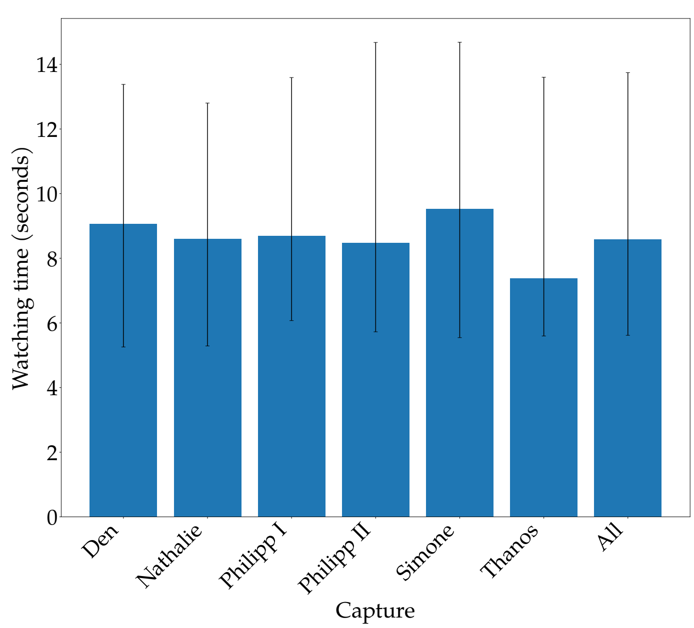
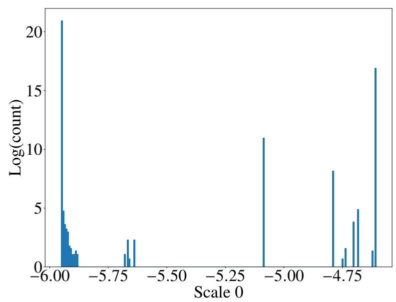
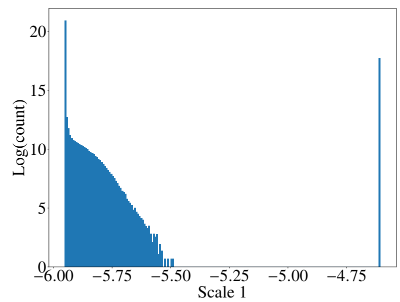


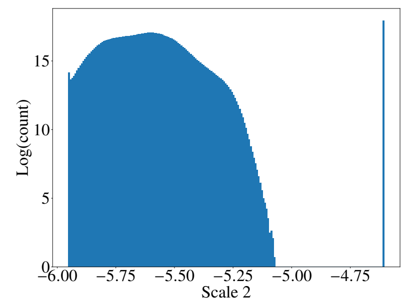
Fig. 4: Bar chart of the per-capture watching times for the GS vs. PC and PC vs. GS stimuli (bars = median, error bars = interquartile range).



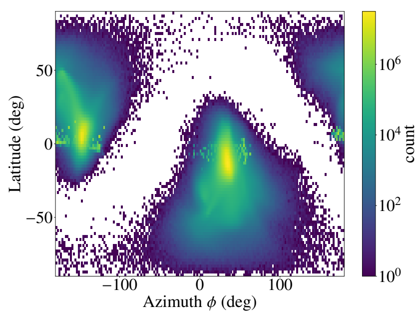
(a) Scale 0 histogram.



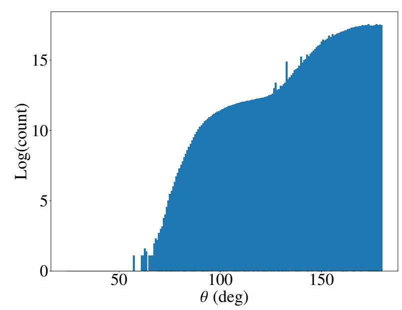
(b) Scale 1 histogram.



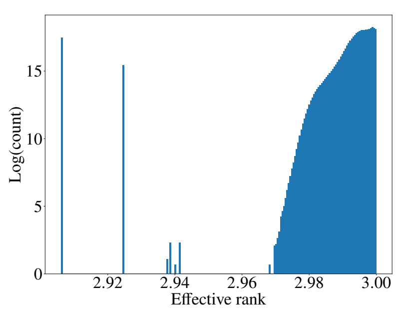
(c) Scale 2 histogram.



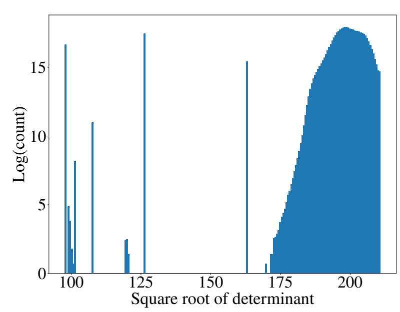
(d) Rotation axis spherical histogram.



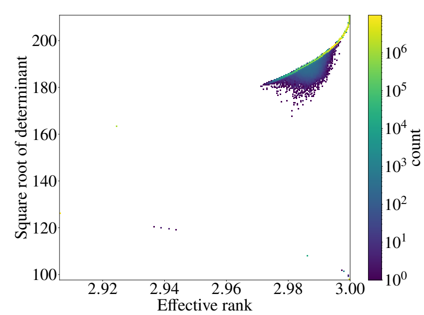
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

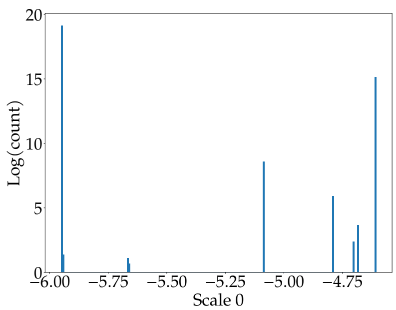


(g) Square root of the covariance matrix determinant.

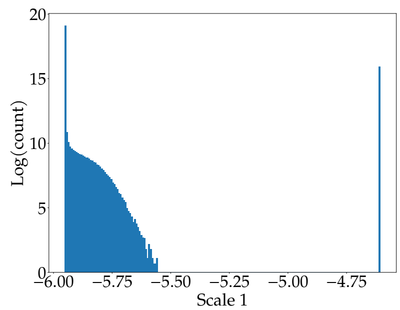


(h) Effective rank vs square root of the covariance matrix determinant.

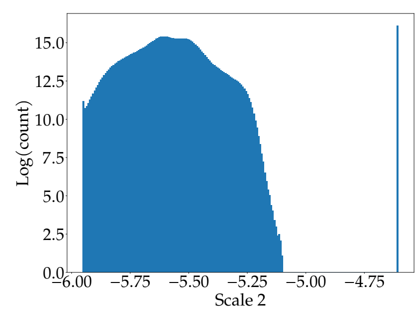
Fig. 5: Gaussian splat analysis of all dynamic reconstructions.



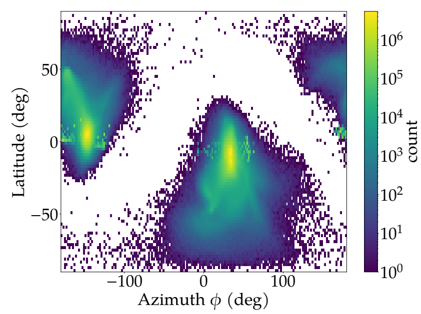
(a) Scale 0 histogram.



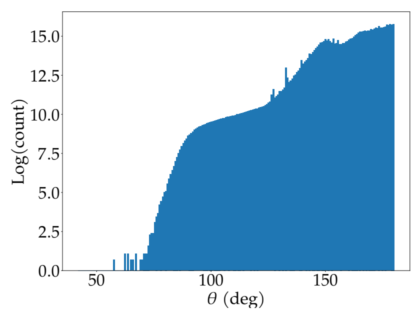
(b) Scale 1 histogram.



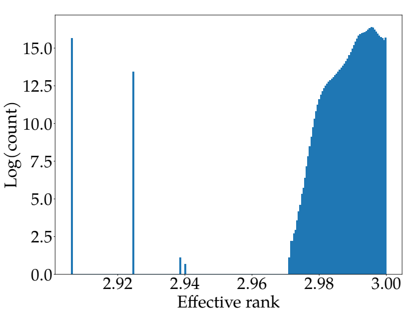
(c) Scale 2 histogram.



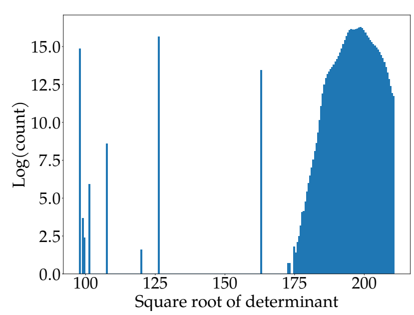
(d) Rotation axis spherical histogram.



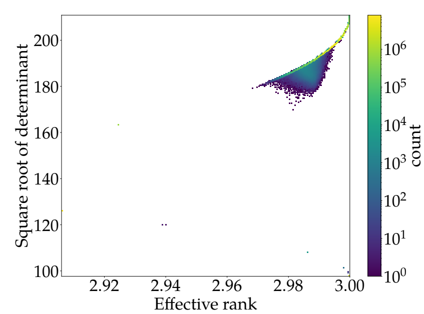
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

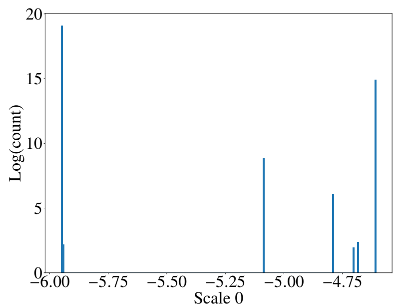


(g) Square root of the covariance matrix determinant.

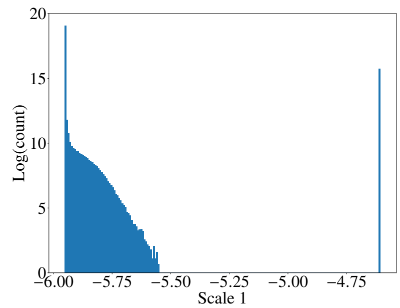


(h) Effective rank vs square root of the covariance matrix determinant.

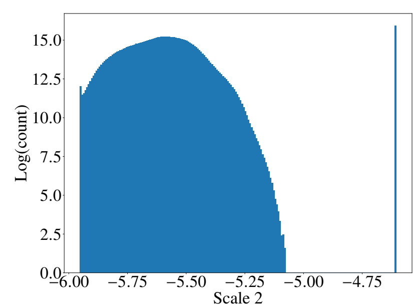
Fig. 6: Gaussian splat analysis of Den's dynamic reconstruction.



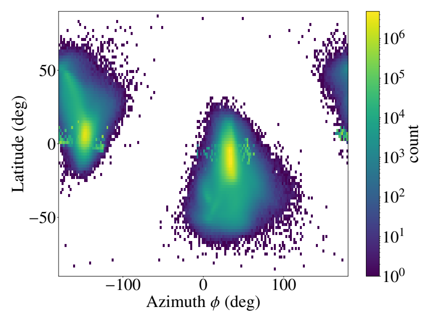
(a) Scale 0 histogram.



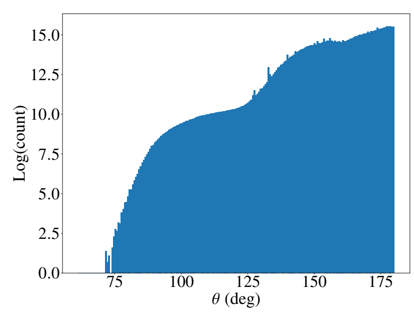
(b) Scale 1 histogram.



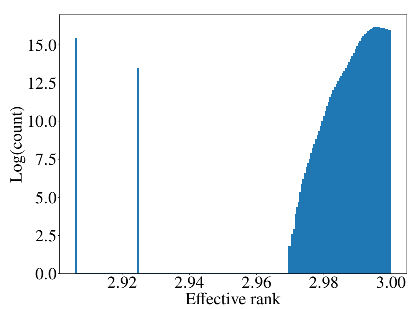
(c) Scale 2 histogram.



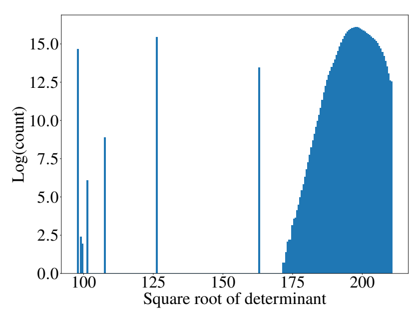
(d) Rotation axis spherical histogram.



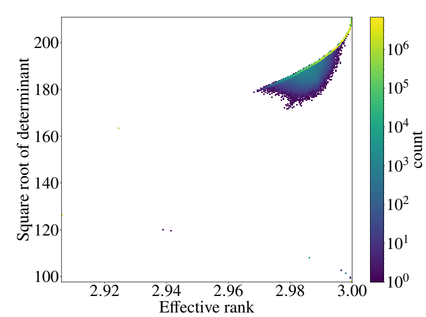
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

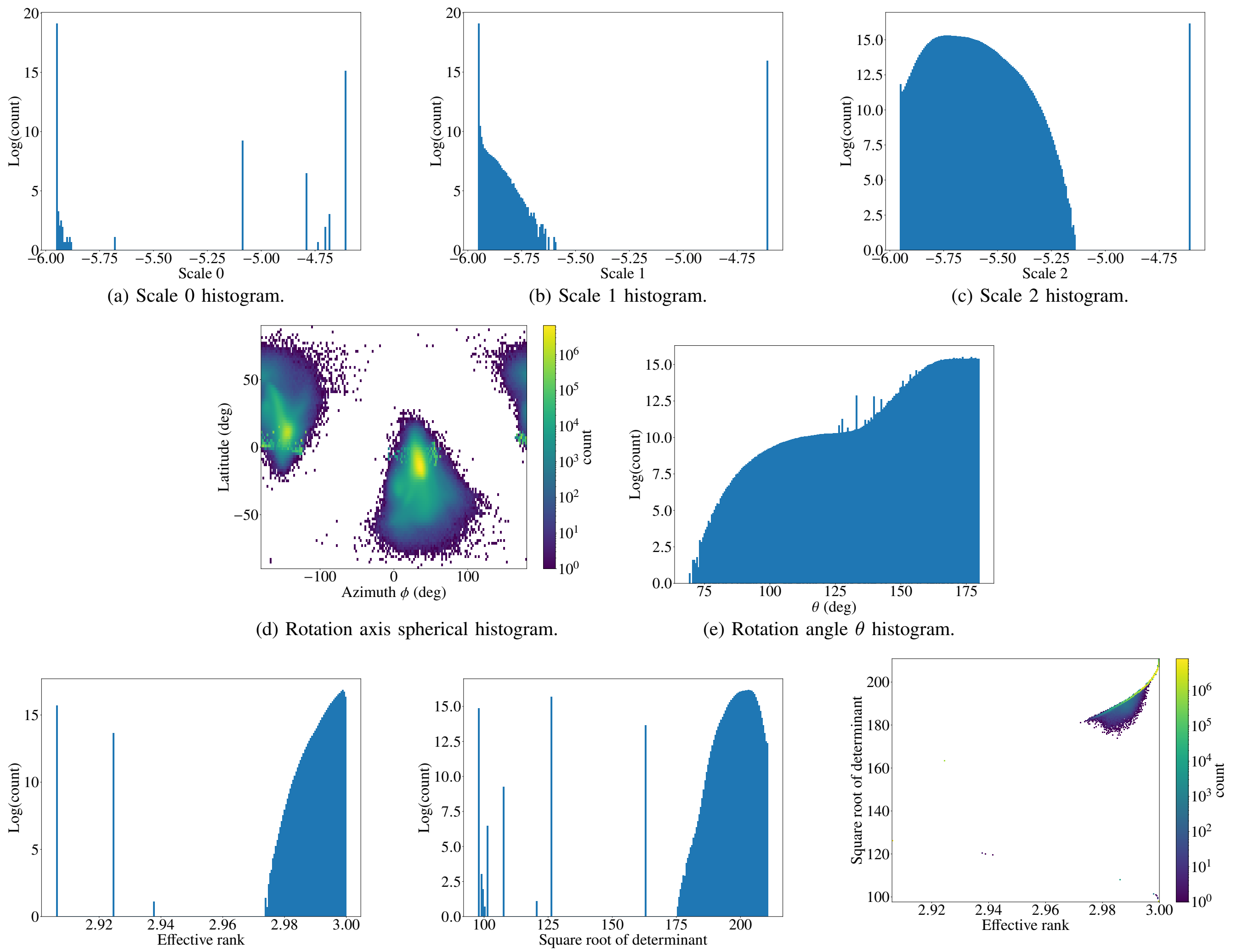


(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 7: Gaussian splat analysis of Nathalie's dynamic reconstruction.



(a) Scale 0 histogram.

(b) Scale 1 histogram.

(c) Scale 2 histogram.

(d) Rotation axis spherical histogram.

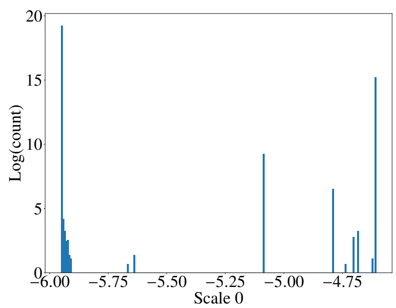
(e) Rotation angle θ histogram.

(f) Covariance matrix effective rank histogram.

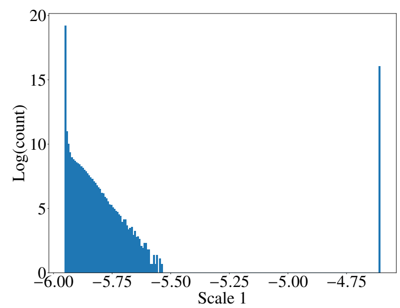
(g) Square root of the covariance matrix determinant.

(h) Effective rank vs square root of the covariance matrix determinant.

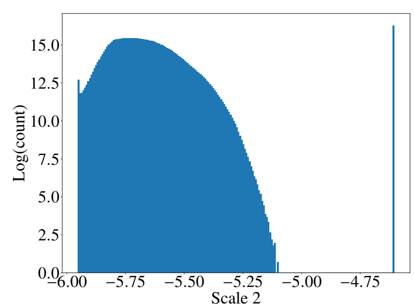
Fig. 8: Gaussian splat analysis of Philipp I's dynamic reconstruction.



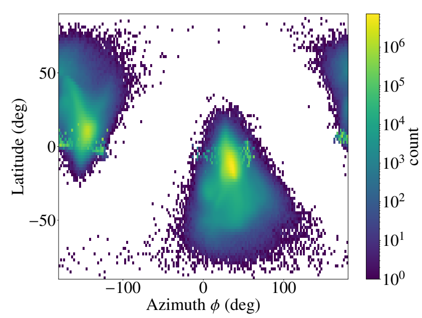
(a) Scale 0 histogram.



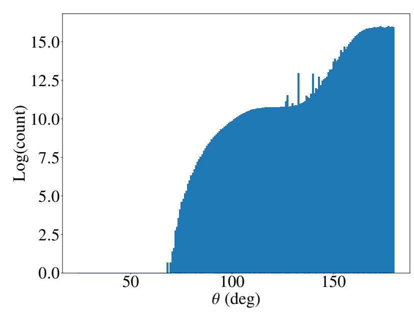
(b) Scale 1 histogram.



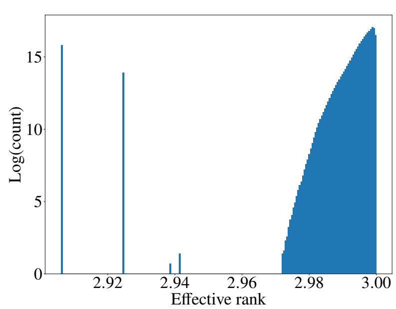
(c) Scale 2 histogram.



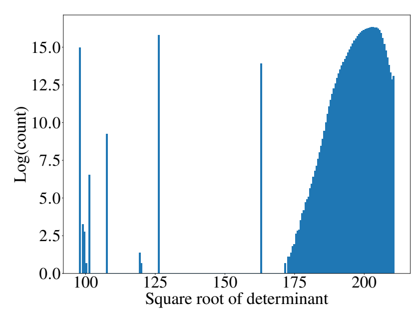
(d) Rotation axis spherical histogram.



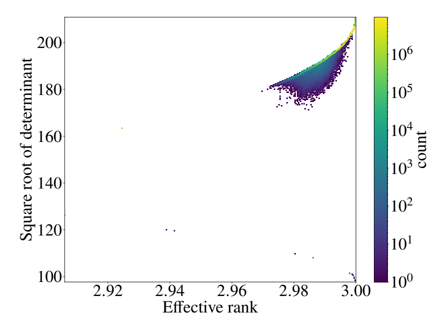
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

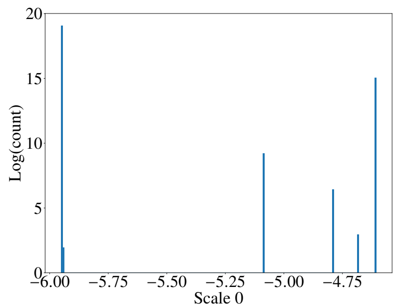


(g) Square root of the covariance matrix determinant.

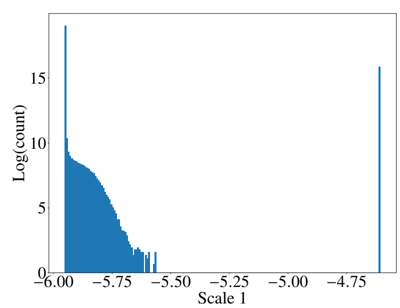


(h) Effective rank vs square root of the covariance matrix determinant.

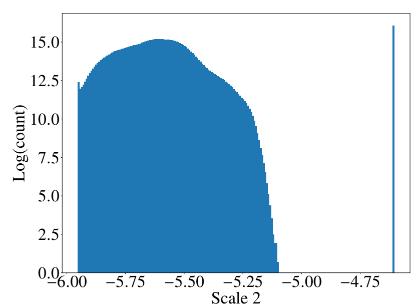
Fig. 9: Gaussian splat analysis of Philipp II's dynamic reconstruction.



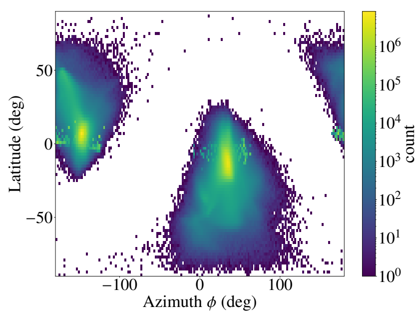
(a) Scale 0 histogram.



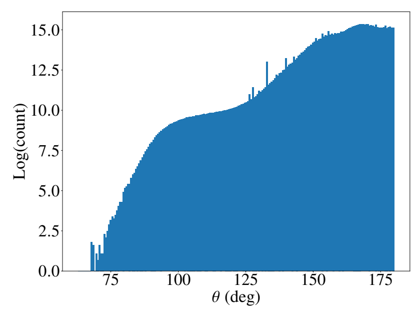
(b) Scale 1 histogram.



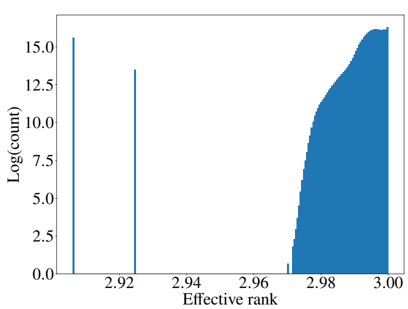
(c) Scale 2 histogram.



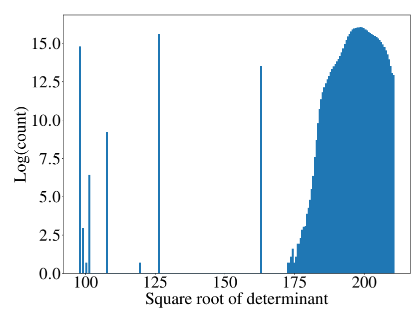
(d) Rotation axis spherical histogram.



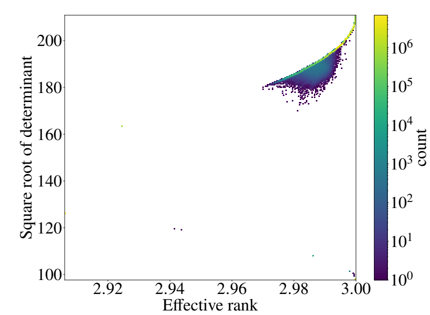
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

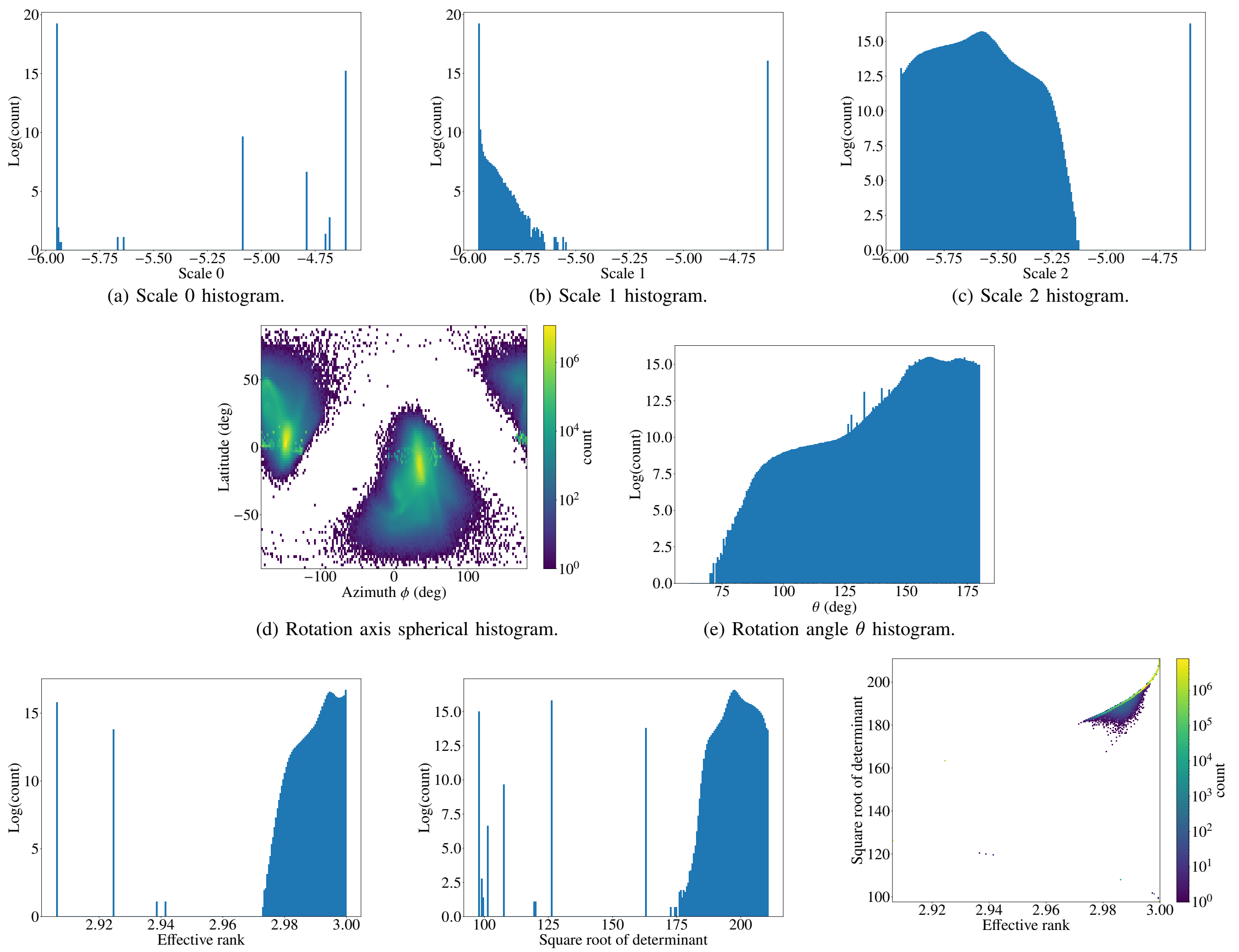


(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 10: Gaussian splat analysis of Simone's dynamic reconstruction.



(a) Scale 0 histogram.

(b) Scale 1 histogram.

(c) Scale 2 histogram.

(d) Rotation axis spherical histogram.

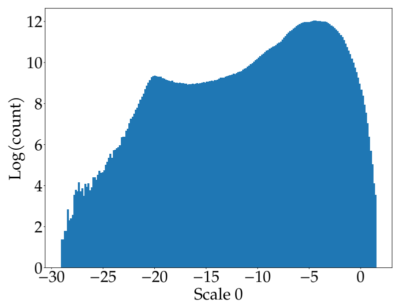
(e) Rotation angle θ histogram.

(f) Covariance matrix effective rank histogram.

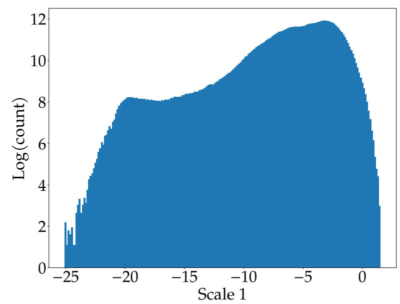
(g) Square root of the covariance matrix determinant.

(h) Effective rank vs square root of the covariance matrix determinant.

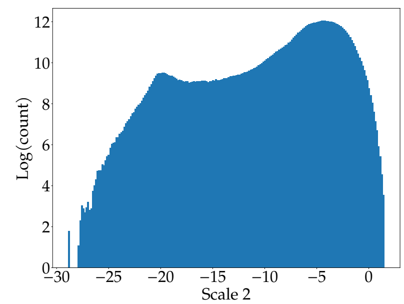
Fig. 11: Gaussian splat analysis of Thanos's dynamic reconstruction.



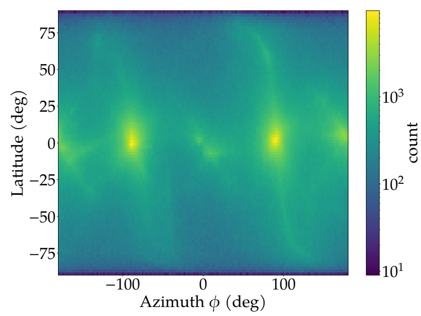
(a) Scale 0 histogram.



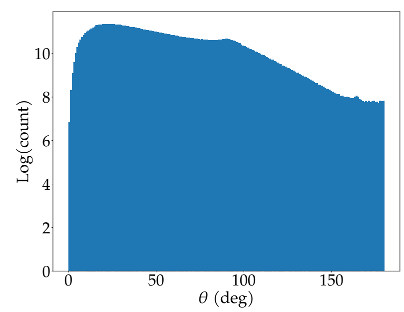
(b) Scale 1 histogram.



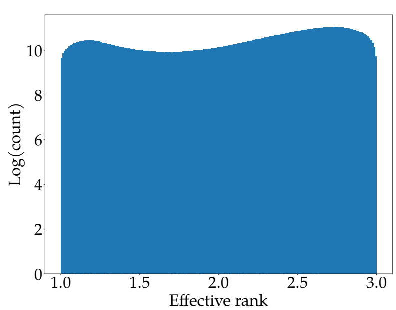
(c) Scale 2 histogram.



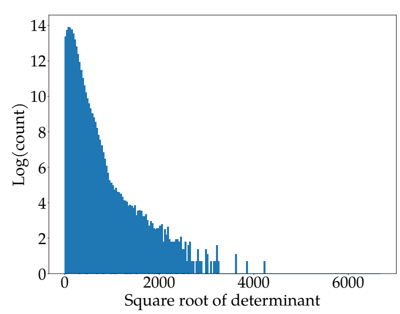
(d) Rotation axis spherical histogram.



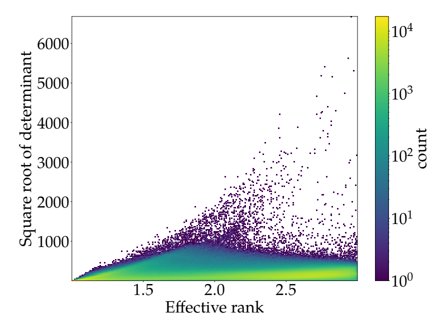
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

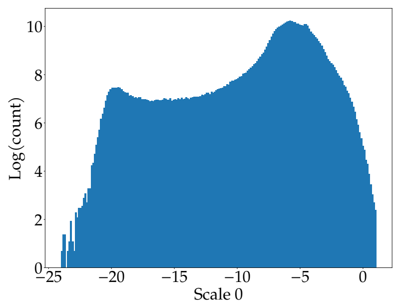


(g) Square root of the covariance matrix determinant.

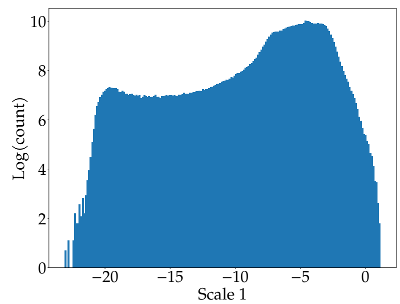


(h) Effective rank vs square root of the covariance matrix determinant.

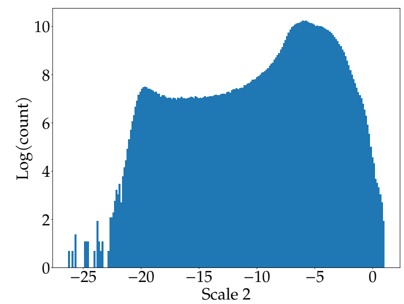
Fig. 12: Gaussian splat analysis of all static reconstructions.



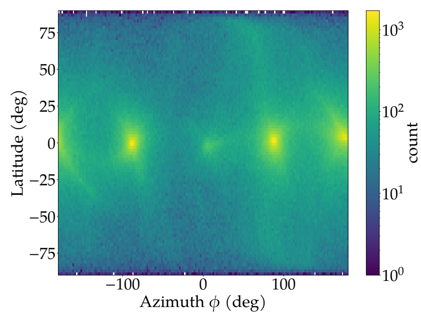
(a) Scale 0 histogram.



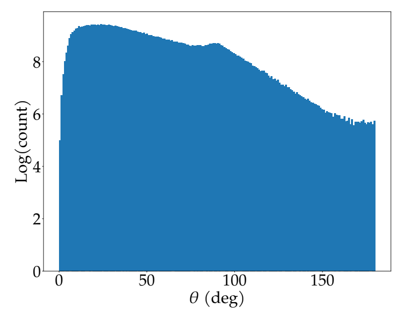
(b) Scale 1 histogram.



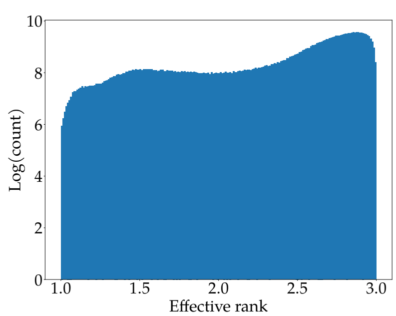
(c) Scale 2 histogram.



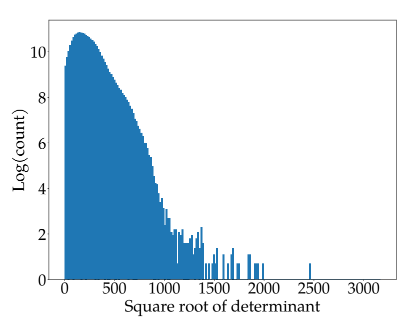
(d) Rotation axis spherical histogram.



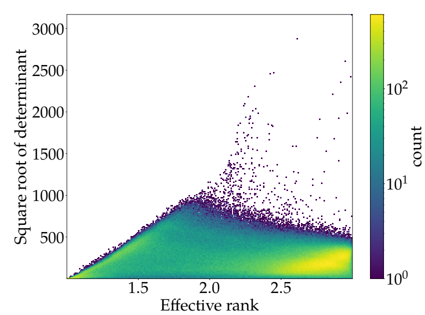
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

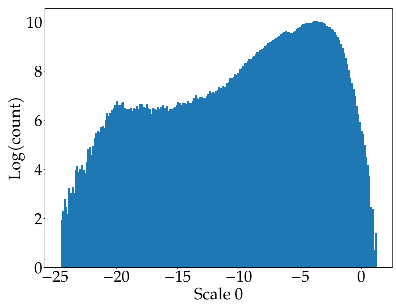


(g) Square root of the covariance matrix determinant.

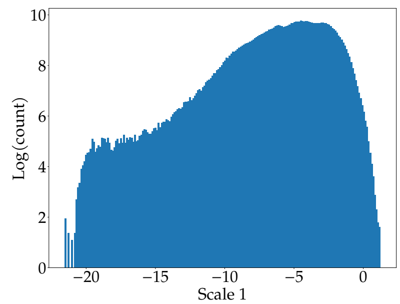


(h) Effective rank vs square root of the covariance matrix determinant.

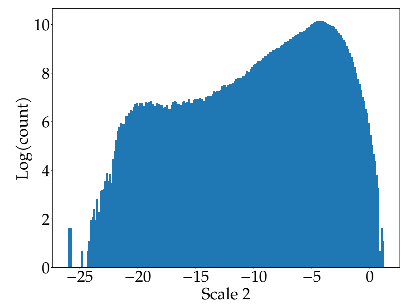
Fig. 13: Gaussian splat analysis of the Chapel static reconstruction.



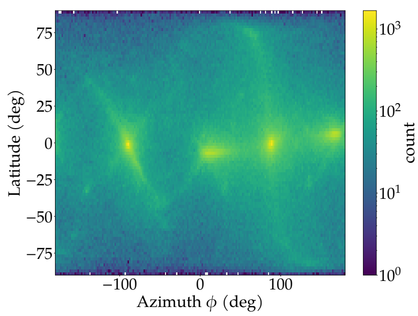
(a) Scale 0 histogram.



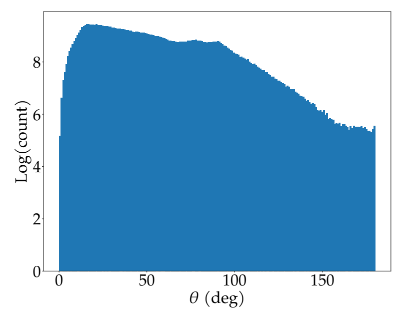
(b) Scale 1 histogram.



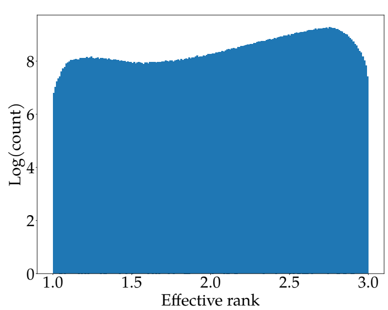
(c) Scale 2 histogram.



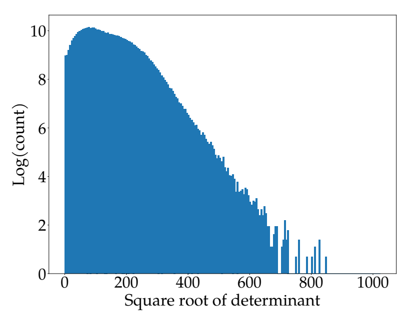
(d) Rotation axis spherical histogram.



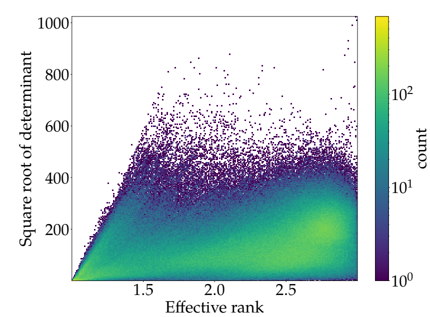
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

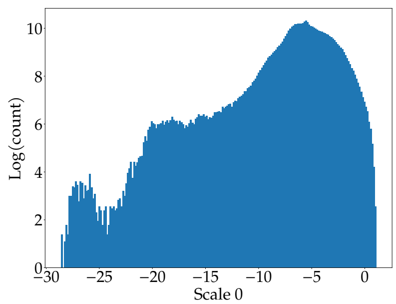


(g) Square root of the covariance matrix determinant.

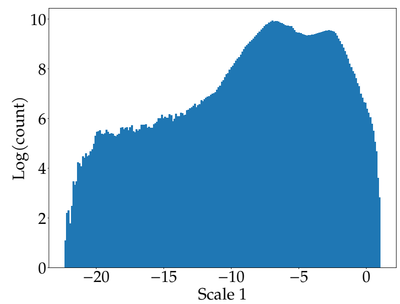


(h) Effective rank vs square root of the covariance matrix determinant.

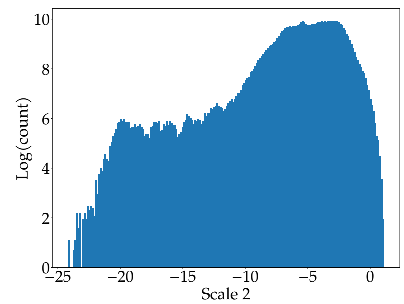
Fig. 14: Gaussian splat analysis of the Lecture Room static reconstruction.



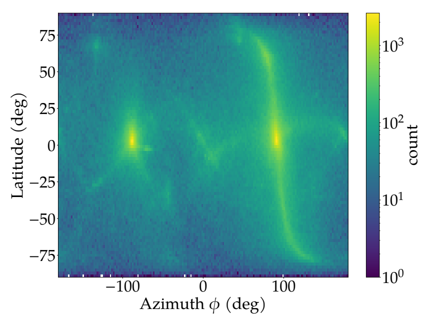
(a) Scale 0 histogram.



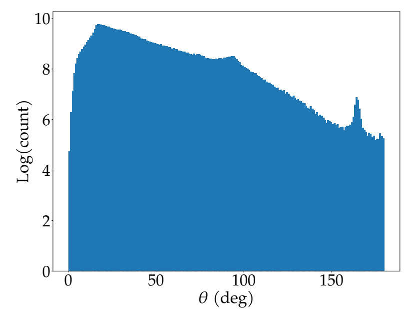
(b) Scale 1 histogram.



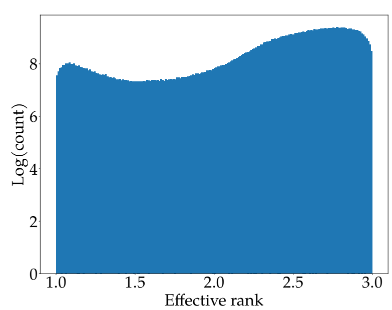
(c) Scale 2 histogram.



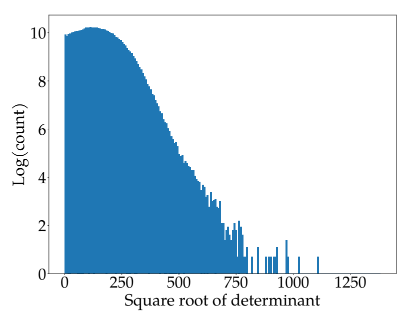
(d) Rotation axis spherical histogram.



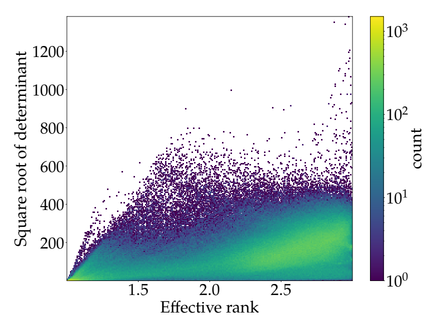
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

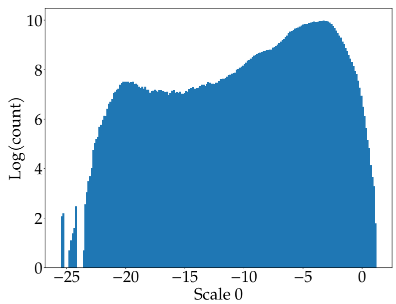


(g) Square root of the covariance matrix determinant.

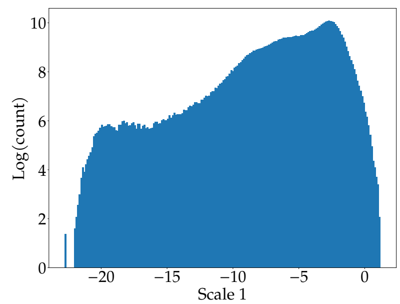


(h) Effective rank vs square root of the covariance matrix determinant.

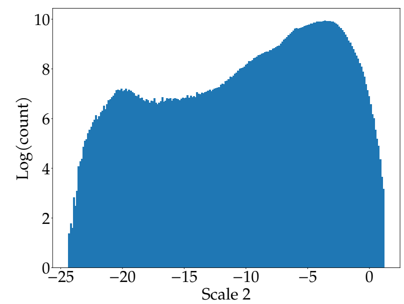
Fig. 15: Gaussian splat analysis of the Lecture Hall static reconstruction.



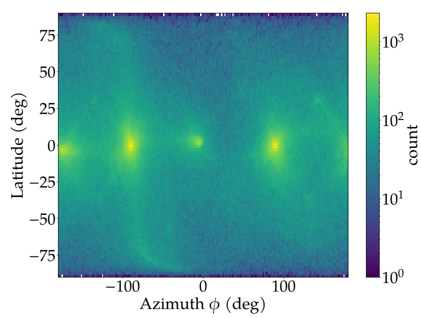
(a) Scale 0 histogram.



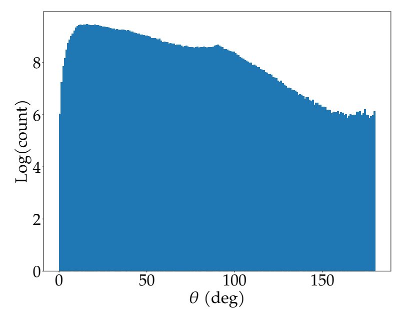
(b) Scale 1 histogram.



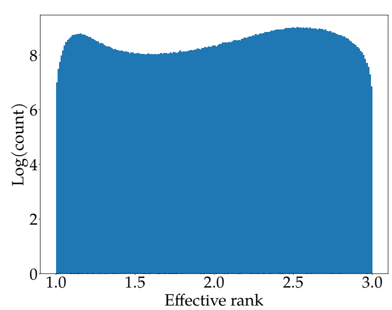
(c) Scale 2 histogram.



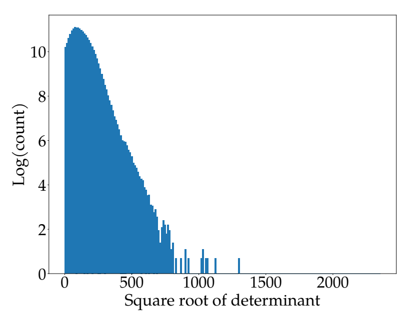
(d) Rotation axis spherical histogram.



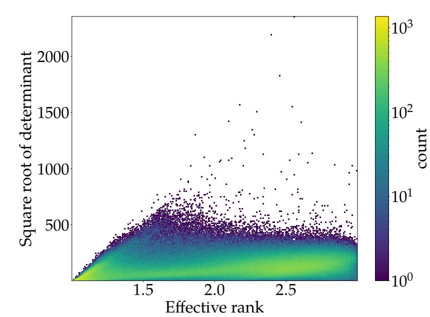
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

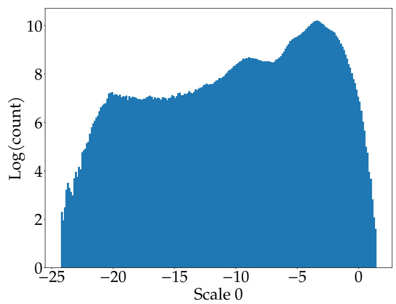


(g) Square root of the covariance matrix determinant.

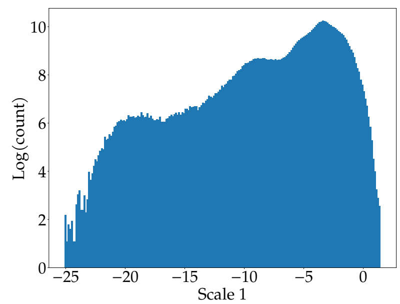


(h) Effective rank vs square root of the covariance matrix determinant.

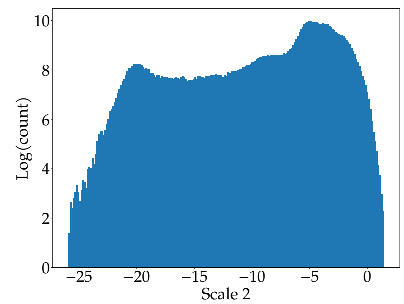
Fig. 16: Gaussian splat analysis of the Meeting Room static reconstruction.



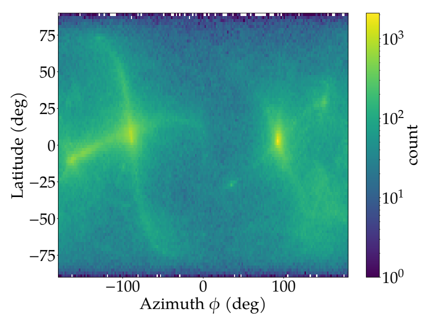
(a) Scale 0 histogram.



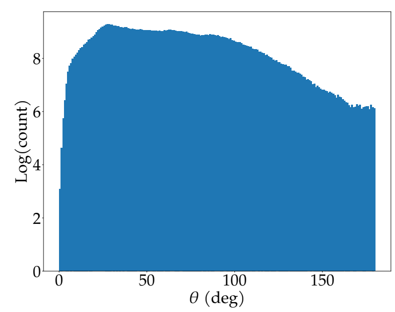
(b) Scale 1 histogram.



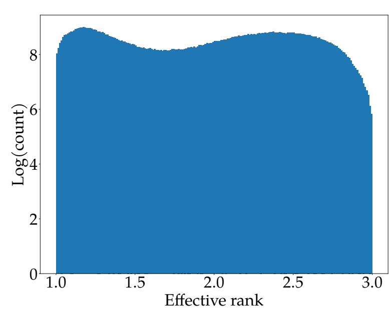
(c) Scale 2 histogram.



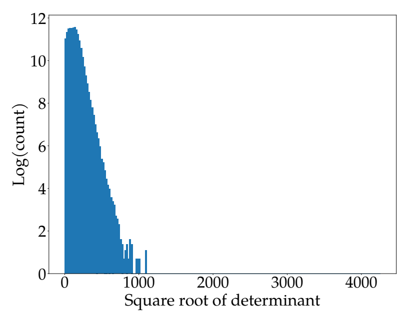
(d) Rotation axis spherical histogram.



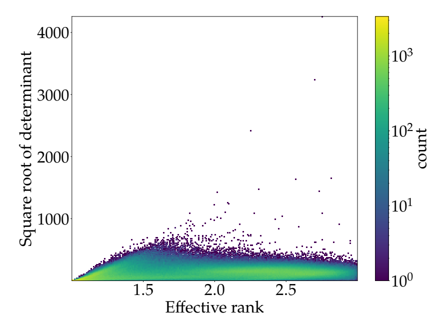
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

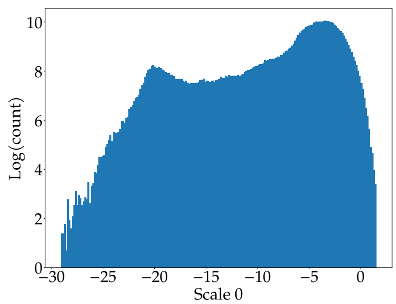


(g) Square root of the covariance matrix determinant.

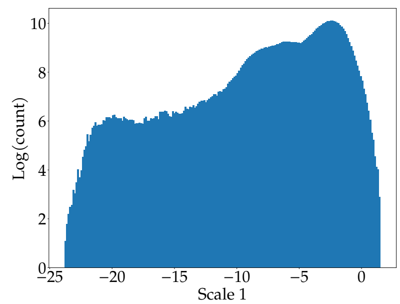


(h) Effective rank vs square root of the covariance matrix determinant.

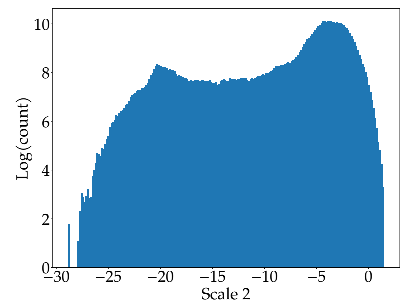
Fig. 17: Gaussian splat analysis of the Office Space static reconstruction.



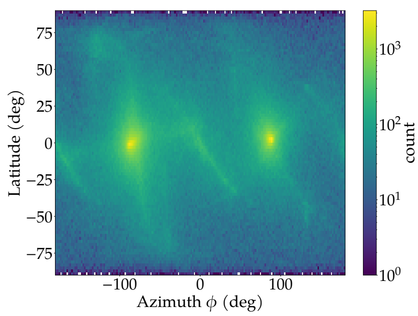
(a) Scale 0 histogram.



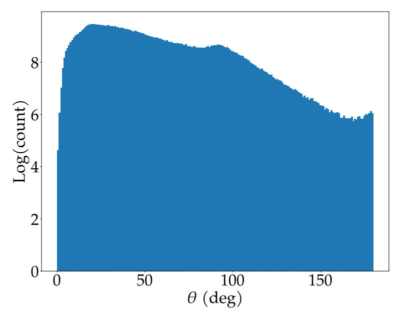
(b) Scale 1 histogram.



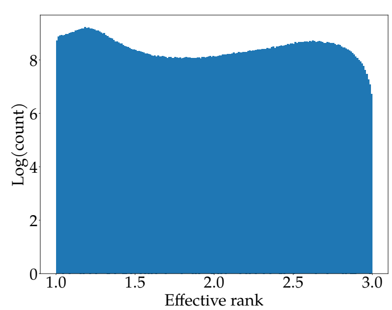
(c) Scale 2 histogram.



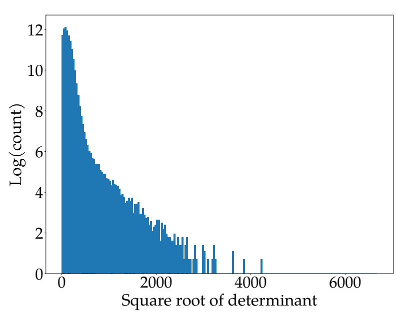
(d) Rotation axis spherical histogram.



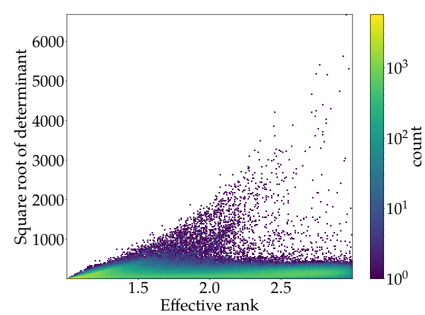
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.

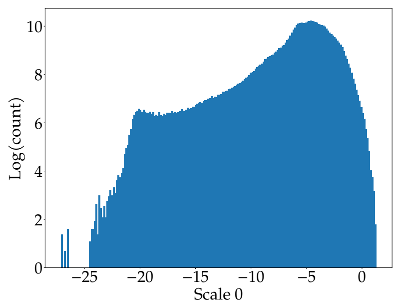


(g) Square root of the covariance matrix determinant.

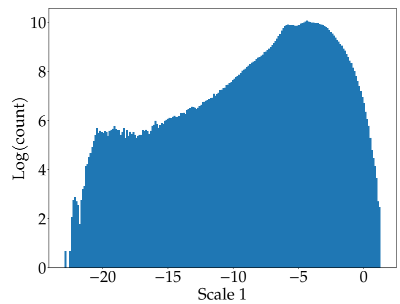


(h) Effective rank vs square root of the covariance matrix determinant.

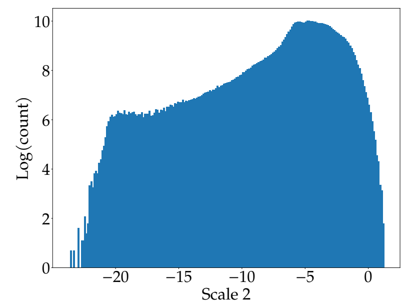
Fig. 18: Gaussian splat analysis of the Outdoor Scene static reconstruction.



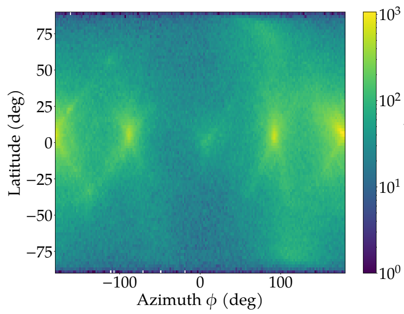
(a) Scale 0 histogram.



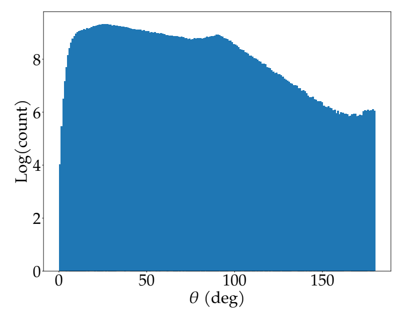
(b) Scale 1 histogram.



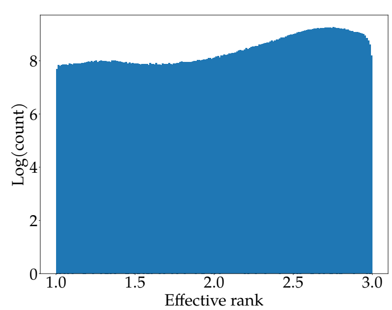
(c) Scale 2 histogram.



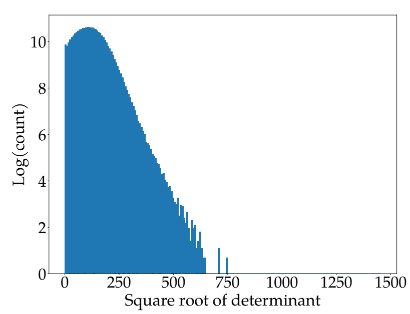
(d) Rotation axis spherical histogram.



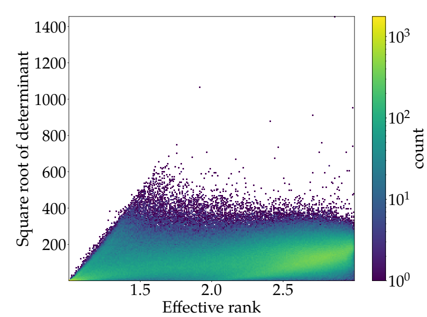
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 19: Gaussian splat analysis of the Cluttered Space static reconstruction.